

Saravanakumar Thirukonda Janarthanan

Full-Stack Software Lead Engineer | Distributed & Multithreaded Systems

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SUMMARY

Full-stack lead engineer with nearly 10 years of experience spanning research and product development. Detail-oriented across the complete product development lifecycle - HLD/LLD, end-to-end feature development, and problem-solving in distributed, multithreaded systems. Hands-on with AI and Generative AI - integrating LLMs and agentic workflows (Model Context Protocol / tool-calling) into engineering and developer-productivity solutions. Proven record of improving resiliency, performance, and code quality while mentoring engineers and driving delivery across geo-distributed teams.

CORE SKILLS

- High-Level (HLD) and Low-Level (LLD) design of complex systems and features.
- Distributed and multithreaded architecture.
- Stakeholder management - driving quarterly/monthly delivery plans.
- Strong data structures and algorithms.
- PaaS containers and cloud-native deployment.
- Reverse-engineering complex, interdependent modules to improve performance and resolve intricate issues.
- Agile methodology with CI/CD.
- Applied AI / GenAI - LLM integration, prompt design, and agentic tooling (MCP / tool-calling).

EXPERIENCE

Visa Bangalore, India

Staff Software Engineer / Team Lead

Aug 2022 - Present

- Lead a team of 4 engineers within the Benefits & Loyalty organization - owning task estimation, sprint planning, mentoring, design/code reviews, on-call resolution, and end-to-end delivery from inception to production.
- Partner with product and strategy teams to understand business features and translate them into technical requirements, roadmaps, and executable engineering plans.
- Designed and delivered GenAI capabilities for the Offer Analysis multi-agent system - integrating LLMs, agentic workflows, and Model Context Protocol (MCP) tooling with a Spring Boot orchestrator and Visa SSO to automate offer analysis and accelerate engineering productivity.
- Applied AI-assisted development (LLM code generation, prompt engineering, and agentic tooling) to speed up delivery and improve code quality across the team.
- Led the design of the probing plane for VDBP (Benefits) across geo-located teams, improving its resiliency and efficiency.
- Independently built React UIs (with unit and end-to-end tests) for the AWS Route 53-based traffic management system and shipped to production - raising code quality by 50%+ and rendering performance by 30%.
- Designed a Kubernetes-based (EKS) automated test-runner to guarantee the integrity of the traffic manager's probing operations.
- Implemented a health system for AWS Route 53 DNS servers and built Docker-based integration suites to verify health-calculation accuracy - increasing code coverage from 50% to 85%.
- Delivered a PoC for automated K8s cluster and application deployment using Terraform and Ansible.
- Led MTE/PROD environment setup and install planning for the GOAT (Global Offers Analytics) platform - namespace provisioning, container product profiles, and Kubernetes environment configuration on CloudView.
- Configured end-to-end platform integrations - S3 connectivity, OpenSearch, SSL/DNS certificates, and SSO onboarding across MTE and PROD environments.
- Authored and executed database provisioning scripts (sequences, tables, materialized views) on IBM DB2 to support GOAT backend and batch services.
- Owned production deployment runbooks and install plans, coordinating cross-team execution for reliable, repeatable releases.
- Versatile across the full stack - frontend (React/Angular), backend (Java/Spring Boot 3, Spring WebFlux reactive, Node.js), data/batch (Kafka, Airflow, Hadoop), and databases (IBM DB2, MySQL, Redis, R2DBC).
- Move fluidly between architecture, hands-on coding, DevOps (Docker, Kubernetes, Jenkins, Terraform/Ansible), and GenAI -

covering design through production operations end to end.

Virtusa Bangalore, India

Technology Consultant Level 3 (Citibank client)

Nov 2020 - Aug 2022

- Worked for Citibank on Anti-Money Laundering (AML) systems, developing and enhancing transaction-monitoring and compliance features.
- Delivered hands-on coding changes and feature improvements to strengthen detection accuracy and regulatory compliance.

Mindtree Bangalore, India

Senior Software Engineer - Level 2 (Bank of America client)

Sep 2019 - Nov 2020

- Worked for Bank of America (BOFA) on the Corporate Card creation platform, building and enhancing card-issuance features.
- Optimized existing workflows and delivered hands-on coding changes to improve reliability and processing efficiency.

Syntel Bangalore, India

Associate Consultant Level 1 (American Express client)

Aug 2017 - Jul 2019

- Joined as a new college graduate (NGC) and worked on American Express's Buyer Initiated Payment (BIP) application.
- Delivered hands-on coding changes and new feature enhancements across the application.

TECHNOLOGY & TOOLS

Languages: Java, Node.js, C#, Ruby on Rails, SQL

Backend & Frameworks: Spring Boot 3, Spring WebFlux (non-blocking/reactive), Spring 6, gRPC, WebSockets

Frontend: React, Angular

Cloud & DevOps: AWS (S3, Route 53), Kubernetes (EKS), Docker, Jenkins, CloudView, Terraform, Ansible

Data & Messaging: Kafka, OpenSearch, Elastic Stack, Hadoop/HDFS, Airflow, AutoSys, Control-M, Sqoop

Databases: IBM DB2, MySQL, Redis, R2DBC

Security & Auth: OAuth 2.0, Visa SSO (ForgeRock)

AI / GenAI: LLM integration (Claude, GPT), multi-agent / agentic workflows, Model Context Protocol (MCP), tool-calling, prompt engineering, RAG, AI-assisted development

Tools & OS: Git, Linux

PROFESSIONAL ACHIEVEMENTS

- Visa Star Award - "Unsung Champion" (FY25 Q1, Dec 2024): delivered the AAA initiative for the VAS Benefits program - migrated the entire application from IBM DB2 to MySQL, segregated the read-only datasets, and automated the end-to-end data migration executed on release day.
- **Promoted to Staff Data Engineer at Visa:** recognized for optimizing the offer rules engine (batch -> real-time offer allotments), leading VDBP- AAA backend restructuring, and establishing OperA alerts and Grafana production monitoring.
- **Mindtree SpotOn "A-Team" awards (2019 & 2020):** on-time completion of the 20.01.00 release despite tight timelines, and exemplary team spirit.
- **Mindtree SpotOn "Perseverance" award (2020):** unrelenting dedication and consistent team collaboration.

EDUCATION

B.Tech, Information Technology

Anna University, Chennai, TN, India

2013 - 2017

Date: 11 Jul 2026

Preferred work location: Bangalore